

This technical report provides the detailed pseudocode of the proposed greedy algorithms and the proofs of the theoretical results. Algorithm 1 (Followers) describes the procedure for computing the follower set induced by an anchored edge. Algorithm 2 (GIA) presents the greedy incremental anchoring strategy, which iteratively selects the edge–weight increment pair with the largest follower gain per unit budget. Algorithm 3 (GPA) accelerates GIA by using candidate reduction, upper-bound pruning, and intermediate-result reuse, while preserving the same greedy selection criterion.

A Pseudo code of Followers

Algorithm 1: Followers(G, e, δ_e, s, C_s)

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1 PQ ← queue(); // sort by coreness-layer ascending order
2  $\mathcal{F} \leftarrow \emptyset$ ;
3  $w(e) \leftarrow w(e) + \delta_e$ ;
4 for each  $x \in \{u, v\}$  if  $x \notin C_s$  do
5   PQ.add( $x$ );
6 while PQ  $\neq \emptyset$  do
7    $x \leftarrow \text{PQ.pop}()$ ;
8    $w^+(x) \leftarrow \text{SupportWeightedDegree}(G, x)$ ;
9   if  $w^+(x) \geq s$  then
10     $\mathcal{F} \leftarrow \mathcal{F} \cup \{x\}$ ;
11    for each  $y \in N(x, G)$  and  $x \prec y$  do
12      if  $y \notin C_s$  and  $y \notin \mathcal{F}$  then
13        PQ.add( $y$ );
14   else
15     Q ← queue();
16     Q.push( $x$ );
17     while Q  $\neq \emptyset$  do
18        $y \leftarrow \text{Q.pop}()$ ;
19       if  $y \in \mathcal{F}$  then
20          $\mathcal{F} \leftarrow \mathcal{F} \setminus \{y\}$ ;
21       for each  $z \in N(y, G) \cap \mathcal{F}$  do
22          $w^+(z) \leftarrow w^+(z) - w(y, z)$ ;
23         if  $w^+(z) < s$  then
24           Q.push( $z$ );
25  $w(e) \leftarrow w(e) - \delta_e$ ;
26 return  $\mathcal{F}$ 

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Algorithm description. Algorithm 1 computes the follower set induced by an anchored edge via a localised simulation of the s -core peeling process. Starting from the anchored endpoints, the algorithm explores only nodes that are reachable through neighbours with higher coreness–layer pairs, following the upstairs paths determined by the onion-layer structure. A node is retained as a follower if its support weighted degree meets the s -core threshold; otherwise, it is discarded. Discarding a node may reduce the support of neighbouring retained followers, potentially triggering further removals. This cascading update exactly mirrors the behaviour of the s -core decomposition, but is restricted to the structurally affected region. The process terminates when no further admissible nodes remain, and the retained follower nodes constitute the final follower set.

B Pseudo code of GIA

Algorithm description. Algorithm 2 presents the baseline greedy incremental anchoring procedure of GIA. Starting from the initial s -core, the algorithm iteratively evaluates candidate edges whose endpoints are not both contained in the current s -core. For each candidate edge, GIA determines the required weight increment using the

Algorithm 2: GIA(G, b, s)

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1  $G' \leftarrow G, A \leftarrow \emptyset, \text{sum} \leftarrow 0$ ;
2  $C_s \leftarrow s\text{-core}(G')$ ;
3 while  $\text{sum} < b$  do
4    $E^* \leftarrow \binom{V}{2} \setminus \binom{C_s}{2}$ ;
5    $e^* \leftarrow \text{none}, \delta^* \leftarrow 0, \text{max\_FR} \leftarrow 0$ ;
6   for each  $e \in E^*$  do
7      $\delta_e \leftarrow \text{computeDelta}(G', s, e)$ ;
8     if  $\text{sum} + \delta_e \leq b$  then
9        $\mathcal{F}(e, \delta_e) \leftarrow \text{Followers}(G', e, \delta_e, s, C_s)$ ;
10       $\text{FR}(e) \leftarrow |\mathcal{F}(e, \delta_e)| / \delta_e$ ;
11      if  $\text{max\_FR} < \text{FR}(e)$  then
12         $e^* \leftarrow e, \delta^* \leftarrow \delta_e$ ;
13         $\text{max\_FR} \leftarrow \text{FR}(e)$ ;
14    $w_{G'}(e^*) \leftarrow w_{G'}(e^*) + \delta^*$ ;
15    $\text{sum} \leftarrow \text{sum} + \delta^*$ ;
16    $A \leftarrow A \cup (e^*, \delta^*)$ ;
17    $C_s \leftarrow s\text{-core}(G')$ ;
18 return  $A$ 

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CGI rule, computes the induced follower set, and evaluates the corresponding follower ratio. At each iteration, the edge achieving the largest follower ratio is selected, its weight is increased accordingly, and the s -core is updated. This process repeats until the available budget is exhausted, yielding a greedy baseline for comparison.

C Pseudo code of GPA

Algorithm description. Algorithm 3 presents the pseudocode of GPA, an accelerated variant of GIA that reduces the candidate space while preserving the same greedy selection criterion. GPA exploits the locality of follower propagation by organising nodes outside the s -core into connected components and maintaining, for each component, the best candidate edge–weight increment pair identified so far. At each iteration, GPA selects the current best candidate across all components. It then explores candidate edges whose endpoints belong to different components, guided by node-level and edge-level upper bounds that allow early termination and safe pruning. Only candidate pairs that cannot be ruled out by these bounds are evaluated using the same follower computation as in GIA. After applying the selected edge reinforcement, GPA updates the component structure. If the reinforced edge connects two distinct components, they are merged and their best candidate is recomputed; otherwise, only the affected component is updated. In both cases, the s -core is recomputed locally within the affected components. This process repeats until the budget is exhausted, yielding the same greedy outcome as GIA with fewer candidate evaluations.

D Proof of Theorem 1

PROOF. We reduce the Maximum Coverage (MC) problem, which is known to be NP-hard unless $P = NP$ [1], to the WASCO problem. The MC problem is defined as follows: given a collection of sets $\{T_1, T_2, \dots, T_t\}$ containing elements from a universe $\{e_1, e_2, \dots, e_d\}$

Algorithm 3: GPA(G, b, s)

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1  $G' \leftarrow G, A \leftarrow \emptyset, \text{sum} \leftarrow 0;$ 
2  $L \leftarrow \text{list}(), M \leftarrow \text{map}();$  // maps each component to its
   best  $(e, \delta_e, FR)$ 
3  $C_s \leftarrow s\text{-core}(G');$ 
4  $M \leftarrow \text{buildComponents}(G', b, s);$ 
5 while  $\text{sum} < b$  do
6    $e^*, \delta^*, \text{max\_FR} \leftarrow \arg \max_{(e, \delta_e, FR) \in M} FR;$ 
7   for  $u \in V \setminus C_s$  and  $\text{sum} + (s - c(u)) \leq b$  do
8      $U(u) \leftarrow \text{UpperBound}(G', u, s);$ 
9      $L.append(u);$ 
10   $\text{sort}(L);$  // by upper bound decreasing order
11  for  $u \in L$  do
12    if  $\text{max\_FR} > 2 \cdot U(u)$  then
13      break;
14    for  $v \in L$  s.t.  $v$  appears after  $u$  do
15      if  $u.\text{component} = v.\text{component}$  then
16        continue;
17      if  $\text{max\_FR} > U(u) + U(v)$  then
18        break;
19      if  $\text{max\_FR} > U(u, v)$  then
20        continue;
21      else
22         $e \leftarrow (u, v);$ 
23         $\delta_e \leftarrow \text{computeDelta}(G', s, e);$ 
24        if  $\text{sum} + \delta_e \leq b$  then
25           $\mathcal{F}(e, \delta_e) \leftarrow \text{Followers}(G', e, \delta_e, s, C_s);$ 
26           $FR(e) \leftarrow |\mathcal{F}(e, \delta_e)| / \delta_e;$ 
27          if  $\text{max\_FR} < FR(e)$  then
28             $e^* \leftarrow e, \delta^* \leftarrow \delta_e;$ 
29             $\text{max\_FR} \leftarrow FR(e);$ 
30   $w_{G'}(e^*) \leftarrow w_{G'}(e^*) + \delta^*;$ 
31   $\text{sum} \leftarrow \text{sum} + \delta^*;$ 
32   $A \leftarrow A \cup (e^*, \delta^*);$ 
33   $\text{updateComponents}(G', b, s, \text{sum}, M);$ 
34 return  $A$ 

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and a budget b' , the objective is to select b' sets such that the number of unique elements covered is maximised.

To construct an instance of the WASCO problem, consider an arbitrary instance I_{MC} of the MC problem with sets $\{T_1, \dots, T_t\}$ and elements $\{e_1, \dots, e_d\} = \bigcup_{i=1}^t T_i$, as illustrated in the top of Figure 1. We define an instance $I_{WASCO} = (G, s, b)$ as follows:

- The graph G consists of three components: a part M representing the sets T_i , a part N representing the elements e_j , and a set of $(d+2)$ -cliques.
- For each set T_i , create a corresponding set of nodes M_i in M , where $M_i = \{u_{i,1}, u_{i,2}, \dots, u_{i,d+2}\}$. The nodes $\{u_{i,1}, \dots, u_{i,d+1}\}$ form a $(d+1)$ -clique, and $u_{i,d+2}$ is connected to $\{u_{i,1}, \dots, u_{i,d}\}$.
- For each element e_j , create a corresponding node v_j in N . Add an edge between v_j and $u_{i,j}$ for every set T_i such that $e_j \in T_i$.

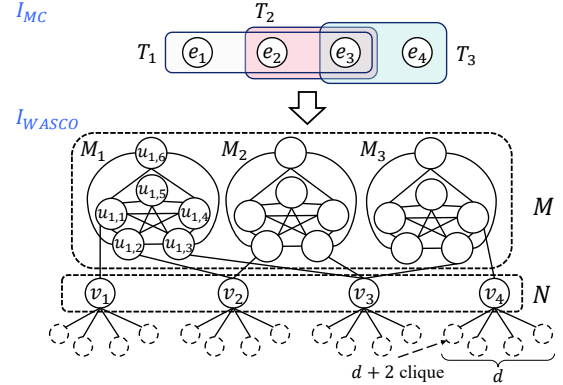


Figure 1: Proof of NP-hardness

- For each v_j , add d disjoint $(d+2)$ -cliques, and connect v_j to one node in each clique.
- Set $s = d+1$ and $b = b'$.
- Every edge in G has a weight of 1.

This construction can clearly be completed in polynomial time. Initially, during the s -core decomposition, all nodes in M and N are removed, leaving only the $(d+2)$ -cliques. This happens because:

- Nodes $u_{i,d+1}$ and $u_{i,d+2}$ in M_i have degree d , which is less than $s = d+1$, and are therefore removed.
- Removing $u_{i,d+1}$ and $u_{i,d+2}$ causes the remaining nodes $u_{i,j}$ in M_i to fail the s -core condition, as their degree drops below s .
- All v_j nodes in N are similarly removed because their connections to M_i are lost.

As a result, only the $(d+2)$ -cliques remain in the s -core after this initial decomposition. To maximise the s -core size, an optimal solution is to add an edge between $u_{i,d+1}$ and $u_{i,d+2}$ in M_i . Adding this edge corresponds to selecting the set T_i in the MC problem. Note that another optimal solution adds edges between distinct sets in M (e.g., $(u_{i,d+2}, u_{j,d+1})$ and $(u_{j,d+2}, u_{i,d+1})$), which can always be replaced by the intra-set connection since both strategies yield the exact same number of followers. When the edge $(u_{i,d+1}, u_{i,d+2})$ is added, this single edge brings the entire M_i together with all of its adjacent v_j nodes in N into the s -core:

- All nodes in M_i are included, since the degrees of $u_{i,d+1}$ and $u_{i,d+2}$ become $d+1$.
- All v_j connected to M_i are included as well, as their degree conditions are now satisfied.

We now argue that this connection is optimal. Our strategy gains an entire M_i and its adjacent v_j nodes at the cost of a single action, so to outperform it, an alternative must achieve a larger gain at the same cost. We therefore compare every candidate from the viewpoint of the gain a single weight increment produces. Recall that M_i survives if and only if *both* $u_{i,d+1}$ and $u_{i,d+2}$ reach a weighted degree of $d+1$; otherwise, once either is removed, every remaining node $u_{i,j}$ with $j \leq d$ also drops below s and the whole M_i collapses. We thus examine every way of spending one unit of weight:

- **Raising the weight of an existing edge from $u_{i,d+1}$ or $u_{i,d+2}$ to a node in $\{u_{i,1}, \dots, u_{i,d}\}$ by 1.** This lifts only one of the two deficient nodes to weighted degree $d+1$, so the other still drops out and all of M_i collapses. The gain is 0.

- **Cross connections between distinct M_i and M_j** (e.g., adding weight on $(u_{i,d+2}, u_{j,d+1})$ and $(u_{j,d+2}, u_{i,d+1})$). This can revive both M_i and M_j , but it spends two units—exactly the same budget as the two intra connections $(u_{i,d+1}, u_{i,d+2})$ and $(u_{j,d+1}, u_{j,d+2})$ —and revives the same M_i , hence the same v_j nodes. It can therefore always be replaced by intra connections with no loss. However, cross connections to a node in $\{u_{\cdot,1}, \dots, u_{\cdot,d}\}$ are inferior for the same reason as the first case.
- **Adding weight within N** (i.e., on (v_j, v_k)). Since each v_j is already connected to d of the $(d+2)$ -cliques, it only needs one more unit of weighted degree to survive; thus a single unit on (v_j, v_k) can bring *two* nodes v_j and v_k into the s -core. However, this gain of 2 is far smaller than the gain of our strategy, so it is not optimal.
- **Connecting a node in N to $u_{i,d+1}$ or $u_{i,d+2}$** . Again only one of the two deficient nodes is filled, so the other collapses and M_i cannot survive. The gain is 0.
- **Adding weight to an edge involving a node in the $(d+2)$ -cliques**. By the same mechanism, this fills at most one deficient node, which is insufficient to revive any M_i . The gain is 0.

Therefore, among all candidates, spending one unit of weight on the connection $(u_{i,d+1}, u_{i,d+2})$ yields the largest gain, so an optimal solution spends each of its b units to complete a distinct M_i . This corresponds exactly to selecting the set T_i in the MC problem. The resulting s -core size gain equals

$$\underbrace{(d+2) \cdot b}_{\text{nodes of the selected } M_i} + |\{v_j : e_j \in \bigcup_{i \in S} T_i\}|,$$

where S is the set of selected M_i with $|S| = b$. The first term is constant for any solution using the full budget, while the second counts the distinct elements covered. Maximising the s -core size thus reduces exactly to maximising the number of distinct covered elements—the objective of the MC problem.

Since the optimal way to maximise the s -core is to add edges corresponding to specific T_i sets so as to include as many e_i elements as possible in the s -core, this process directly mirrors the objective of the MC problem to select sets that maximise the coverage of elements. Thus, selecting M_i that are more densely connected to uncovered v_j aligns with the optimisation goals of the MC problem.

Therefore, the MC problem is reducible to the WASCO problem, implying that $I_{MC} \leq_P I_{WASCO}$. As the MC problem is NP-hard, this proves the NP-hardness of the WASCO problem. $\square$

E Proof of Theorem 2

PROOF. Suppose that f is submodular. Then, for any two sets A and B , it must hold that $f(A) + f(B) \geq f(A \cup B) + f(A \cap B)$. Consider a graph consisting of four nodes $\{u, v, x, y\}$ and two edges (u, x) and (v, y) , each with a weight of 1. Let $s = 2$. If $A = \{(u, v), 1\}$ and $B = \{(x, y), 1\}$, $f(A) + f(B) = 0 < f(A \cup B) + f(A \cap B) = 4$. $\square$

F Proof of Property 1

Ties in follower ratio are broken by a fixed total order $\prec_{tb}$ on node pairs (e.g., lexicographic order of vertex identifiers), chosen independently of candidate enumeration. Both GIA and GPA use $\prec_{tb}$, so the greedy winner in any fixed graph state is unique.

PROOF. We show by induction that GPA and GIA select the same edge-weight increment pair at every iteration.

Base case. Both algorithms start from the same input (G, b, s) , compute the same initial s -core $C_s(G)$, the same coreness-layer order, and the same CGI value for every candidate edge. Hence they enter the first iteration with the same state.

Inductive step. Assume that the two algorithms have selected identical edge-increment actions through the first $t-1$ iterations. Then they enter iteration t with the same current graph G_t , the same s -core $C_s(G_t)$, the same coreness-layer order, the same remaining budget, and the same CGI value for every candidate edge. Let e^* be the edge selected by GIA in this state under $\prec_{tb}$. We show that GPA also selects e^* with the same increment δ_{e^*} .

- (1) **Candidate reduction.** If both endpoints of e^* lie outside $C_s(G_t)$, e^* is unaffected and retained. Otherwise, write $e^* = (u, v)$ with $u \notin C_s(G_t)$, $v \in C_s(G_t)$. By Theorem 4, every candidate (u, x) with $x \in C_s(G_t)$ shares the CGI value, follower set, and hence follower ratio of (u, v) , since these depend only on the non- s -core endpoint u . These candidates form one tie-equivalent class. Because GPA keeps the representative of this class according to the same tie-breaking order $\prec_{tb}$, the retained representative is the same edge that GIA would select from this class. Therefore candidate reduction does not alter the greedy choice.
- (2) **Upper-bound pruning.** GPA prunes a candidate only when the current best ratio ρ *strictly* exceeds a valid upper bound on its follower ratio. The partner-level termination, candidate skipping, and global termination rules use the bounds $U(u, v)$, $U(u) + U(v)$, and $2U(u)$, respectively; by Definition 13 and the descending order of $U(\cdot)$, each bound is valid for every candidate it discards. If e^* were pruned with bound $B(e^*)$, then $FR(e^*) \leq B(e^*) < \rho$, meaning GPA already evaluated a candidate beating e^* —contradicting that e^* is the greedy winner. Hence e^* is never pruned. Moreover, because the pruning condition uses a strict inequality, candidates tying the current best are retained, so the common tie-breaking order remains applicable.
- (3) **Intermediate-result reuse.** By Theorem 3, every follower of an anchored edge is reachable from one of its non- s -core endpoints by an upstairs path, so newly engaged nodes stay within the non- s -core components incident to that edge. A component K outside this region is unchanged: a node $x \in K$ could only change its support, coreness-layer pair, or upstairs reachability by becoming adjacent to a newly engaged node, but such a node was non- s -core just before reinforcement and would then have shared x 's component—a contradiction. GPA thus reuses a component's cached best only when the component is unaffected and the cached candidate is still feasible under the remaining budget; otherwise it recomputes. We track e^* under this scheme, in the two ways its endpoints can lie. If both endpoints of e^* lie in the same non- s -core component, then e^* is an intra-component candidate of that component. If the component is recomputed, e^* is evaluated directly. If its cached best is reused, then since the component is unchanged every local candidate keeps its CGI value, follower ratio, and $\prec_{tb}$ -priority, and any internal tie is resolved by the same $\prec_{tb}$; hence the cached best is exactly the edge GIA would select there, namely e^* . Either way e^* reaches the final comparison.

If the endpoints of e^* lie in different components, then e^* is an inter-component candidate, which reuse never caches away: it is freshly evaluated and enters the final comparison directly.

In that comparison, GPA weighs the cached intra-component bests against the freshly evaluated inter-component candidates under the same follower-ratio criterion, the same upper-bound pruning, and the same $\prec_{tb}$. This is precisely the setting of case (ii): e^* is the greedy winner, so it is never pruned and, on any tie, wins under $\prec_{tb}$. Therefore e^* is present and selected.

In every case e^* survives and is selected with the same weight increment δ_{e^*} , so both algorithms enter iteration $t + 1$ in the same state. By induction, GPA and GIA select identically at every iteration and return the same final s -core. $\square$

G Additional Experiment

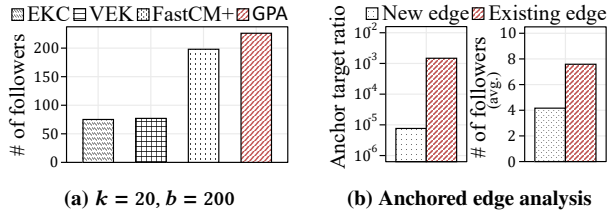


Figure 2: EQ1-4. Effectiveness on the Facebook

EQ1-4. Effectiveness (unweighted graph). We further evaluate GPA on the unweighted Facebook dataset [2] ($|V| = 4,039$, $|E| = 88,234$, $d_{avg} = 43.7$). All compared methods, including EKC [4], VEK [5], and FastCM+ [3], are originally designed for unweighted graphs. To enable a fair comparison, we treat the unweighted graph as a special case of our weighted formulation by assigning unit weight to every edge, and follow the same parameter settings ($k = 20$, $b = 200$) as in FastCM+. As shown in Figure 2(a), GPA can achieve the largest number of followers among all compared methods under the same budget. This result indicates that, even when restricted to the unweighted setting, GPA is capable of operating effectively in practice. Figure 2(b) further analyses the anchoring behaviour of GPA. After normalising by the size of the corresponding feasible action spaces, edge reinforcement is selected more frequently than new edge insertion. Moreover, reinforcement actions tend to yield a higher average number of followers per anchored edge. These observations suggest that strengthening existing connections can be a more effective anchoring strategy than edge insertion, even when all edges have uniform weight.

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